05/06/26 :- Correlation Analysis (Nominal Data) (Numerical data)

χ^2 (chi-square) test for independence :-

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{Expected}}$$

H_0 - Null hypothesis → correlated.
H_a - Alternate hypothesis

If A has c no of values & B has r no of values

then $$\chi^2 = \sum_{i=1}^{c} \sum_{j=1}^{r} \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

$$e_{ij} = \frac{\text{Count}(A = a_i) \times \text{Count}(B = b_j)}{n}$$

Degree of freedom = $(c-1)(r-1)$.

① Suppose that group of 1500 ppl were surveyed. Gender of each person was noted. Each person was tolled was as to whether each of his/her preffered reading material was fiction or non-fiction.

	M	F	
Fiction	250	200	450
Non-fiction	50	1000	1050
	300	1200	1500

H_0 = Gender & preffered reading are independent

H_1 = Gender & preffered reading are dependent.

$$e_{00} = \frac{\text{Count(male)} * \text{count(fiction)}}{n}$$

$$= \frac{300 * 450}{1500} = 90.$$

$$e_{01} = \frac{\text{Count(female)} * \text{count(fiction)}}{n}$$

$$= \frac{1200 * 450}{1500} = \underline{\underline{360}}$$

$$e_{10} = \frac{\text{Count (male)} * \text{count (nonfiction)}}{n}$$

$$= \frac{300 * 1050}{1500} = \underline{\underline{210}}$$

$$e_{11} = \frac{\text{Count (female)} * \text{count (nonfiction)}}{n}$$

$$= \frac{1200 * 1050}{1500} = \underline{\underline{840}}.$$

$$X^2 = \frac{(250-90)^2 + (200-360)^2 + (50-210)^2 + (1000-840)^2}{90 + 360 + 210 + 840}$$

$$= \frac{25600}{90} + \frac{25600}{360} + \frac{25600}{210} + \frac{25600}{840}$$

$$= 284.44 + 71.11 + 121.90 + 30.48$$

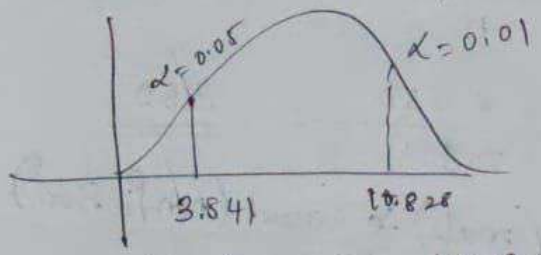
$$= 507.93$$

Degree of freedom = (2-1)(2-1)
= 1

Significance level $\alpha = 0.05$

= 3.841

for $\alpha = 0.01 \Rightarrow 10.828$



507.93 > 10.828 or 3.84.

Null hypothesis is rejected.

ie, Gender & prefered reading are dependent

Mean :-

$$\text{Mean}(x) = \bar{x} = \frac{1}{m}\sum_{i=1}^{m} x_i$$

$$\text{Median}(x) = \begin{cases} x_{(r+1)} & \text{if } m = \text{odd},\ m = 2r+1 \\ \frac{1}{2}(x_r + x_{r+1}) & \text{if } m = \text{even} \\ & m = 2r. \end{cases}$$

$$\text{Variance}(x) = S_x^2 = \frac{1}{m-1}\sum_{i=1}^{m}(x_i - \bar{x})^2$$

$$\text{Standard Deviation} = \sigma = \sqrt{S_x^2} = \sqrt{\frac{1}{m-1}\sum_{i=1}^{m}(x_i - \bar{x})^2}$$

{35, 22, 25, 23, 28, 33, 30}

Mean $(\bar{x}) = \dfrac{35+22+25+23+28+33+30}{7}$

$= 28$

Median $(\tilde{x})$ = 22, 23, 25, 28, 30, 33, 35

$= 28$

$m = 2r+1$

$7 = 2r+1$

$6/2 = r$

$3 = r$

since m is odd, $x_{(r+1)} = x_{(3+1)}$

$= x_4 = 28$

22, 23, 25, 28, 30, 33, 35. $\quad m = 7$.

Mean

$\bar{x} = 28$.

Variance $S_x^2 = \dfrac{1}{m-1} \sum_{i=1}^{m} (x_i - \bar{x})^2$

$= \dfrac{1}{(7-1)} \sum_{i=1}^{7} (22-28)^2 + (23-28)^2 + (25-28)^2 + (28-28)^2 + (30-28)^2 + (33-28)^2 + (35-28)^2$

$= \dfrac{6^2 + 5^2 + 3^2 + 0^2 + 2^2 + 5^2 + 7^2}{6}$

$= \dfrac{148}{6} = 24.66$

σ (SD) $= \sqrt{S_x^2} = \sqrt{\text{variance}} = \sqrt{24.66}$

$= 4.96$

$$\text{Covariance}(x,y) = \frac{1}{n-1}\sum (x_i - \bar{x}) * (y_i - \bar{y})$$

$$\text{Covariance matrix} = \begin{bmatrix} \text{var}(x) & \text{cov}(x,y) \\ \text{cov}(x,y) & \text{var}(y) \end{bmatrix}$$

x_1	x_2
3	7
2	4

$\bar{X}_1 = 5/2 = 2.5$ $\quad \bar{X}_2 = 11/2 = 5.5$

$$\text{Var}(x_1) = \frac{1}{m-1}\sum_{i=1}^{m} (x_i - \bar{x}_1)^2$$

$$= \frac{1}{(2-1)}\sum_{i=1}^{2} (3-2.5)^2 + (2-2.5)^2$$

$$= 0.25 + 0.25 = 0.5$$

$$\text{Var}(x_2) = \frac{1}{1}\sum_{i=1}^{2} (7-5.5)^2 + (4-5.5)^2$$

$$= (1.5)^2 + (1.5)^2$$

$$= 2.25 + 2.25 = 4.5$$

$$\text{Cov}(x_1, x_2) = \frac{1}{2-1}\left(\overset{x_{11}-\bar{x}_1}{3-2.5}\right) * \left(\overset{x_{21}-\bar{x}_2}{7-5.5}\right) + \left(\overset{x_{12}-\bar{x}_1}{2-2.5}\right) * \left(\overset{x_{22}-\bar{x}_2}{4-5.5}\right)$$

$$= (0.5 * 1.5) + (0.5 * 1.5)$$

$$0.75 + 0.75$$

$$= 1.5$$

Cov Matrix (Covariance) $= \begin{bmatrix} 0.5 & 1.5 \\ 1.5 & 4.5 \end{bmatrix}$

08/06/26. **Handling Redundancy in Data Integration**

How to identify :—

→ Correlation analysis

→ Covariance analysis

Redundance possible when multiple DB's are integrated.

→ Same attribute may have diff names in diff dB.

→ One attribute may be a "derived" attribute in another table.

Correlation Analysis :- (χ^2 test or chi-Square test):

Larger χ^2 ⇒ more likely, variables are related.

Correlation Co-efficient – also called Pearson's product moment co-efficient.

$$r_{A,B} = \frac{\sum_{i=1}^{n}(a_i - \bar{A})(b_i - \bar{B})}{(n-1)\sigma_A \sigma_B} = \frac{\sum_{i=1}^{n}(a_i b_i) - n\bar{A}\bar{B}}{(n-1)\sigma_A \sigma_B}$$

n — no of tuples
$\bar{A}$ & $\bar{B}$ — means of A, B.
σ_A & σ_B — SD of A & B.
$\Sigma(a_i b_i)$ — sum of AB cross product.

$r_{A,B} = 0$ = independent
$r_{A,B} < 0$ — negatively correlated
$r_{A,B} > 0$ = +vely correlated (A's values increase as B's).
Higher, stronger correlation.

Visually evaluating Correlation:— Using scatter plots

Correlation viewed as linear relationship:
(diff formulas).

Covariance (Numeric Data). It is similar to variance

$$Cov(A,B) = E((A-\bar{A})(B-\bar{B})) = \frac{\sum_{i=1}^{n}(a_i-\bar{A})(b_i-\bar{B})}{n}$$

Correlation Coefficient: $r_{A,B} = \frac{Cov(A,B)}{\sigma_A \sigma_B}$

+ve covariance:- If $Cov_{A,B} > 0$

$$Cov(A,B) = E(A \cdot B) - \bar{A}\bar{B}.$$

(2,5), (3,8), (5,10), (4,11), (6,14) ⇒ 2 stocks A & B.

PCA ⇒
Data
We
These

07/6/24

Z-Score:

Mean Absolute Deviation = S_A

$$S_A = \frac{1}{n}\left(|V_1 - \bar{A}| + |V_2 - \bar{A}| + |V_3 - \bar{A}| + \ldots + |V_n - \bar{A}|\right)$$

$$V_i' = \frac{V_i - \bar{A}}{S_A}$$

Normalization using decimal scaling.

$$V_i' = \frac{V_i}{10^j} \text{ where } \max(|V_i'|) \leq 1$$

ex:- If A = −986 to 917,

then $\frac{986}{1000} = 0.98 < 1$ $\quad \therefore j = 3$ since $10^3 = 1000$

Q. Given data :— V_1 V_2 V_3 V_4 V_5

200, 300, 400, 600, 1000.

a. MinMax Normalization by setting min = 0, max = 1
b. Z score Normalization
c. Z Score Normalization using Mean Absolute Value.
d. Normalization using decimal scaling.

a. $$V_i' = \frac{V_i - \min_A}{\max_A - \min_A}(\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

$\text{new_min}_A = 0, \quad \text{new_max}_A = 1$

$$200' = \frac{200 - 200}{1000 - 200}(1 - 0) + 0 = 0$$

$300' = \frac{300-200}{1000-200}(1-0)+0 = \frac{100}{800} = \frac{1}{8}$

$= 0.125$

$400' = \frac{400-200}{1000-200}(1-0)+0 = \frac{200}{800} = \frac{1}{4} = 0.25$

$600' = \frac{600-200}{1000-200}(1-0)+0 = \frac{400}{800} = \frac{1}{2} = 0.5$

$1000' = \frac{1000-200}{1000-200}(1-0)+0 = \frac{800}{800} = 1$

So the new set :- $A' = \{0, 0.125, 0.25, 0.5, 1\}$

These values are normalized now.

2. Z-Score normalization :-

$v_i' = \frac{v_i - (\mu_A \text{ or } \bar{A})}{\sigma_A}$ where μ = mean & σ = SD

Mean = $200+300+400+600+1000/5 = 500$

$(x-\bar{A})^2$ or μ^2
$(200-500)^2 = 300^2$
$(300-500)^2 = 200^2$
$(400-500)^2 = 100^2$
$(600-500)^2 = 100^2$
$(1000-500)^2 = 500^2$

$\sigma_A = \sqrt{\frac{\Sigma}{n}(x_i-\mu)^2}$

$\sigma_A = \sqrt{\frac{1}{5}(300^2+200^2+100^2+100^2+500^2)}$

$= \sqrt{80000} = 282.84$

v_1' for 200 :- $\frac{200-500}{282.84} = -1.06$

v_2' for 300 :- $\frac{300-500}{282.84} = -0.707$

v_3' for 400 :- $\frac{400-500}{282.84} = -0.35$

v_4' for 600 :- $\frac{600-500}{282.84} = 0.354$

v_5' for 1000 :- $\frac{1000-500}{282.84} = 1.77$

$v_i = \{200, 300, 400, 600, 1000\}$

$v_i' = \{-1.06, -0.707, -0.35, +0.354, +1.77\}$

3. Z-Score Normalization using Mean Absolute Value

$$v_i' = \frac{v_i - \bar{A}}{SA}$$

$$SA = \frac{1}{n}\left(|v_1 - \bar{A}| + \{|v_2 - \bar{A}| + |v_3 - \bar{A}| + \cdots + |v_n - \bar{A}|\right)$$

$$= \frac{1}{5}\left(|200-500| + |300-500| + |400-500| + |600-500| + |1000-500|\right)$$

$$= \frac{1}{5}(1200) = 240$$

$$v_1' = \frac{200-500}{240} = -1.25$$

$$v_2' = \frac{300-500}{240} = -0.833$$

$v_3^1 = \frac{400-500}{240} = -0.4166$

$v_4^1 = \frac{600-500}{240} = +0.417$

$v_5^1 = \frac{1000-500}{240} = 2.08$

$v_i = \{200, 300, 400, 600, 1000\}$

$v_i^1 = \{-1.25, -0.833, -0.417, +0.417, +2.08\}$

d. Normalization by decimal scaling :-

$v_i^1 = \frac{v_i}{10^j}$ where $\max |v_i^1| \leq 1$

$\max \{1000\}$ $j = 3$

$\max\{2, 3, 4, 6, 10\} \; 10^3 = 1000$

$v_1^1 = \frac{200}{10^3} = 0.2$

$v_2^1 = \frac{300}{10^3} = 0.3$

$v_3^1 = \frac{400}{10^3} = 0.4$

$v_4^1 = \frac{600}{10^3} = 0.6$

$v_5^1 = \frac{1000}{10^3} = 1.0$

$v_i^1 = \{0.2, 0.3, 0.4, 0.6, 1.0\}$

Discretization

4, 8, 15, 21, 21, 24, 25, 28, 34 → Freq = 3

1st sort, then apply if not sorted

* ~~Partition into equisized bin~~
  a. Equal frequency
  b. Smoothing by bin mean.
  c. Smoothing by bin boundary.

a. equal freq:-

Bin 1 – 4, 8, 15
Bin 2 – 21, 21, 24
Bin 3 – 25, 28, 34

b. Smoothing by bin mean:

~~Mean = 4+8+15+21+21+24+25+28+34/9~~

Bin 1 mean = 4+8+15/3 = 27/3 = 9

Bin 2 mean = 21+21+24/3 = 66/3 = 22

Bin 3 mean = 25+28+34/3 = 87/3 = 29.

Bin 1: 9, 9, 9.
Bin 2: 22, 22, 22
Bin 3: 29, 29, 29

c. Smoothing by bin boundary

Bin 1 – 4, 4, 15
Bin 2 – 21, 21, 24
Bin 3 – 25, 25, 34

4 is near to 4
8 is near to 4
15 is near to 15
Similarly for others.

② 5, 10, 11, 13, 15, 35, 50, 56, 72, 92, 204, 215

Total = 12, frequency = 4.

So 3 bins.

a. Equal freq :-

bin 1 :- 5, 10, 11, 13

bin 2 :- 15, 35, 50, 56.

bin 3 : 72, 92, 204, 215

b. Smoothing by bin mean.

bin 1 :- 39/4 = 9.75.

bin 2 = 156/4 = 114

bin 3 :- 583/4 = 145.75

bin 1 : 9.75, 9.75, 9.75

bin 2 : 114, 114, 114

bin 3 : 145.75, 145.75, 145.75

c. Smoothing by bin boundary

bin 1 : 5, 13, 13, 13

bin 2 : 15, 55, 56, 56.

bin 3 : 72, 72, 215, 215

10/06/26 Apriori Algorithm — for pattern mining

↓

It gives frequent pattern